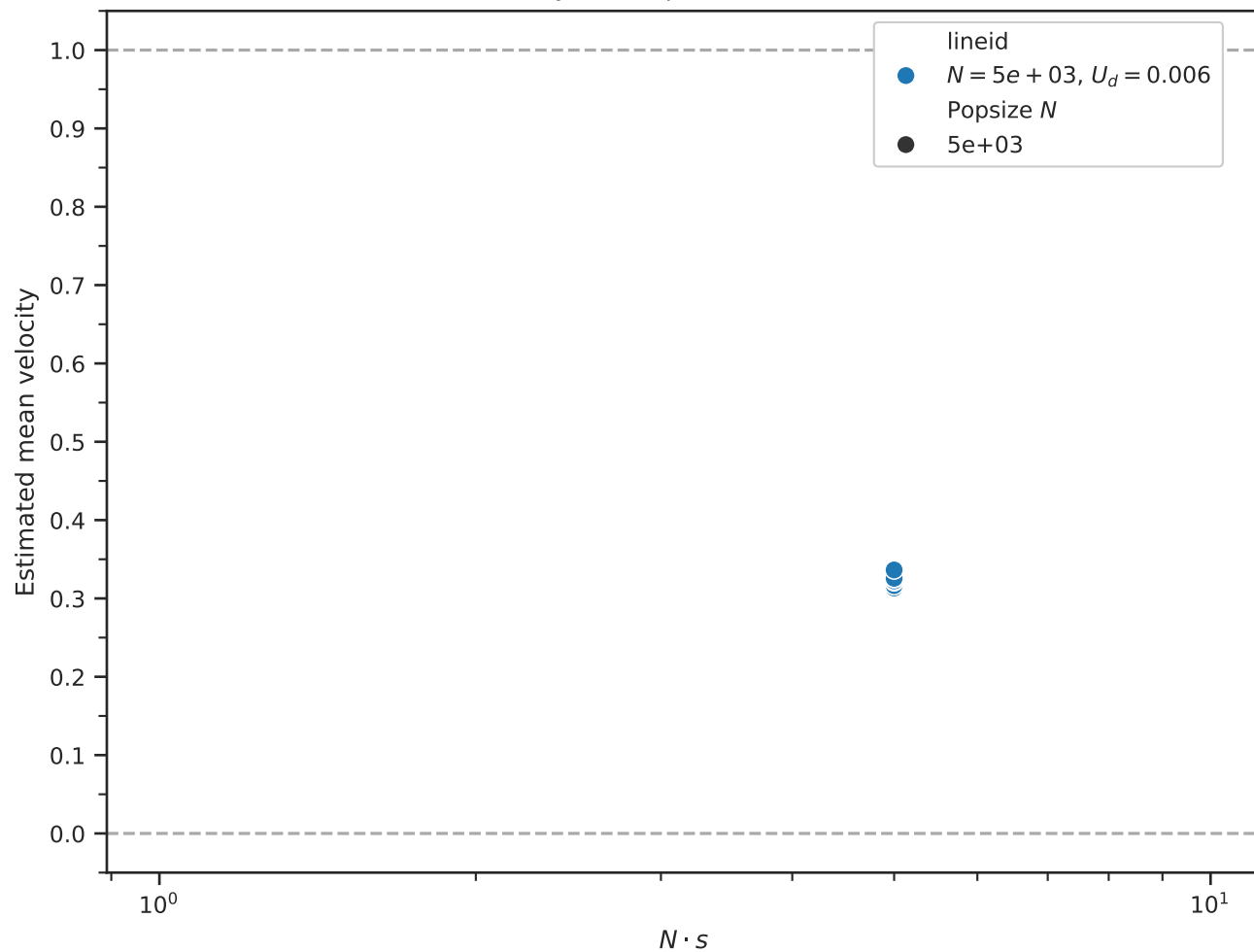
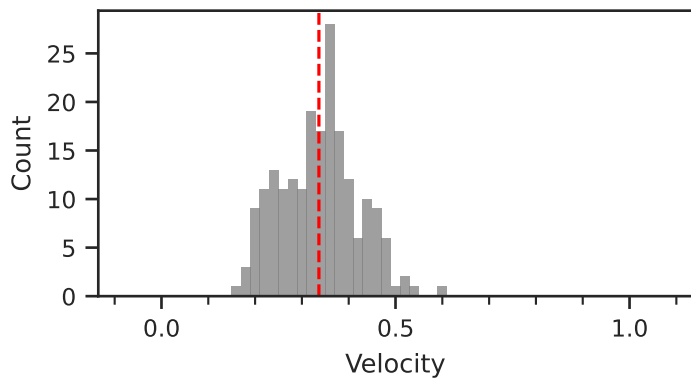


Mean velocity across parameter combinations

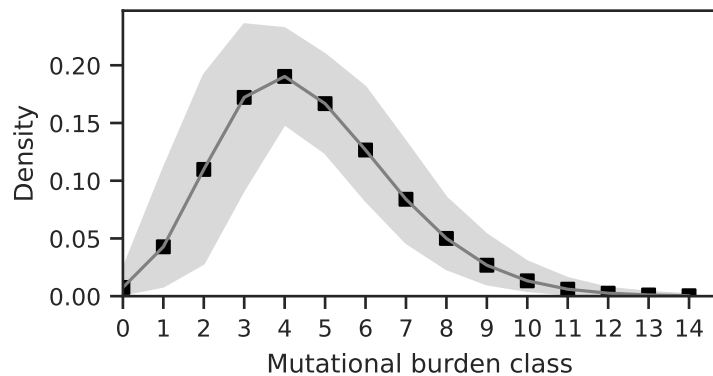


$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 2.5 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3364 ± 0.0822 , $n = 200$ simulations

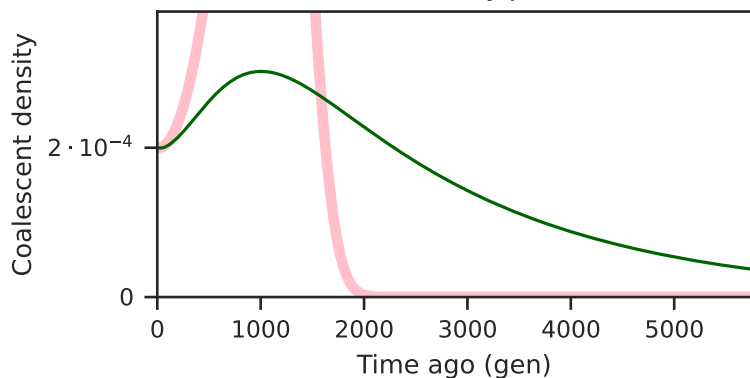
A Velocity Distribution



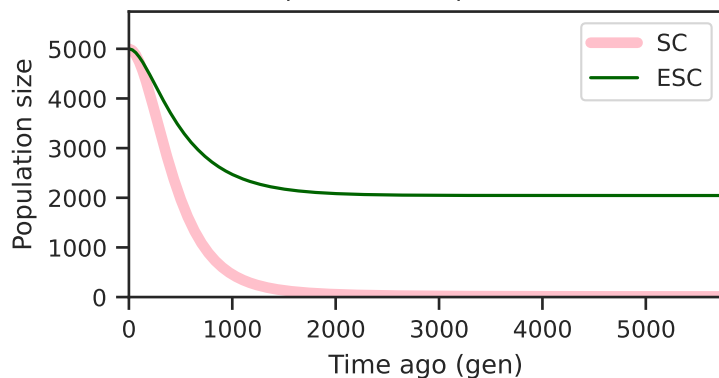
B Mutational burden profile



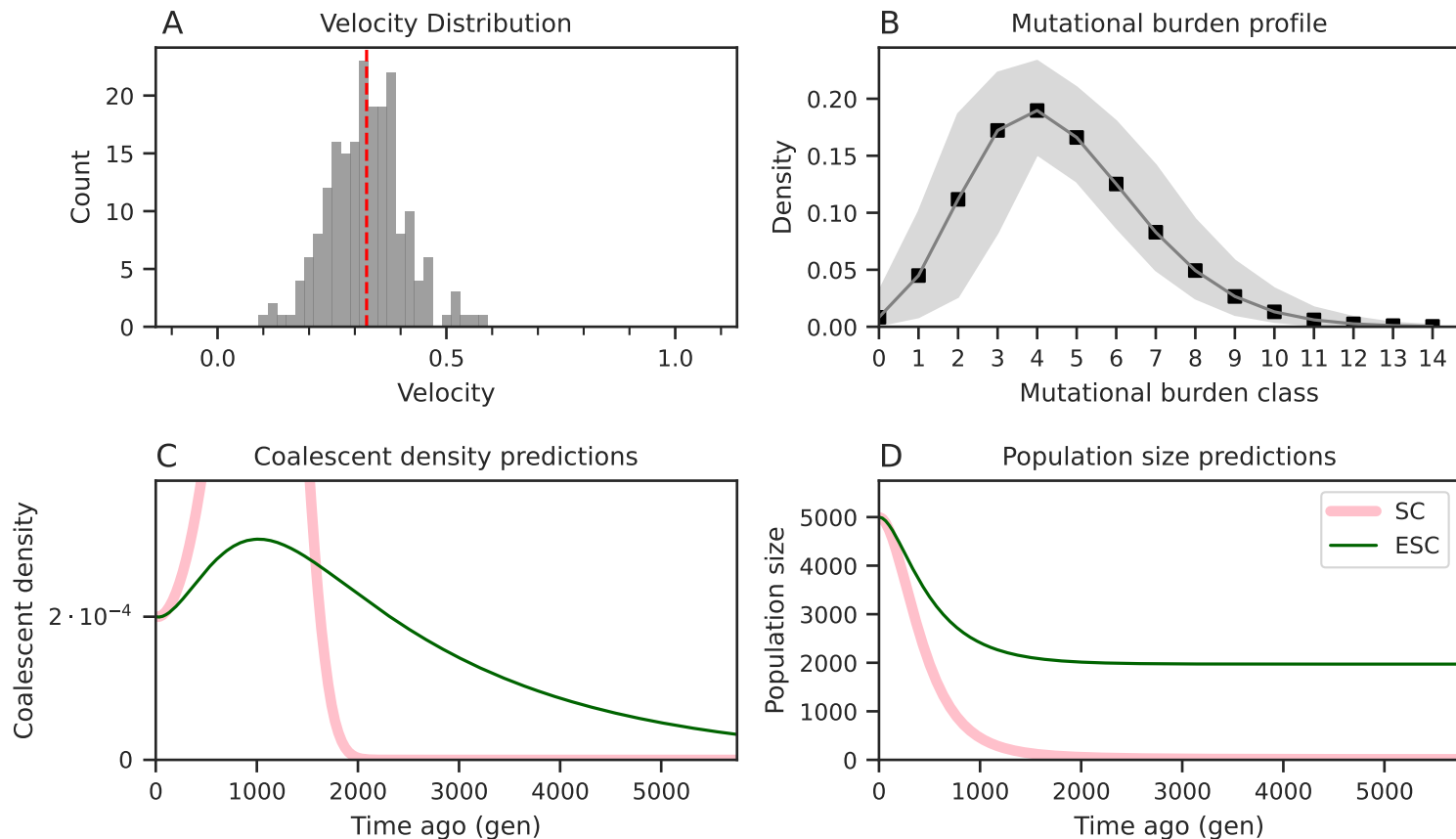
C Coalescent density predictions



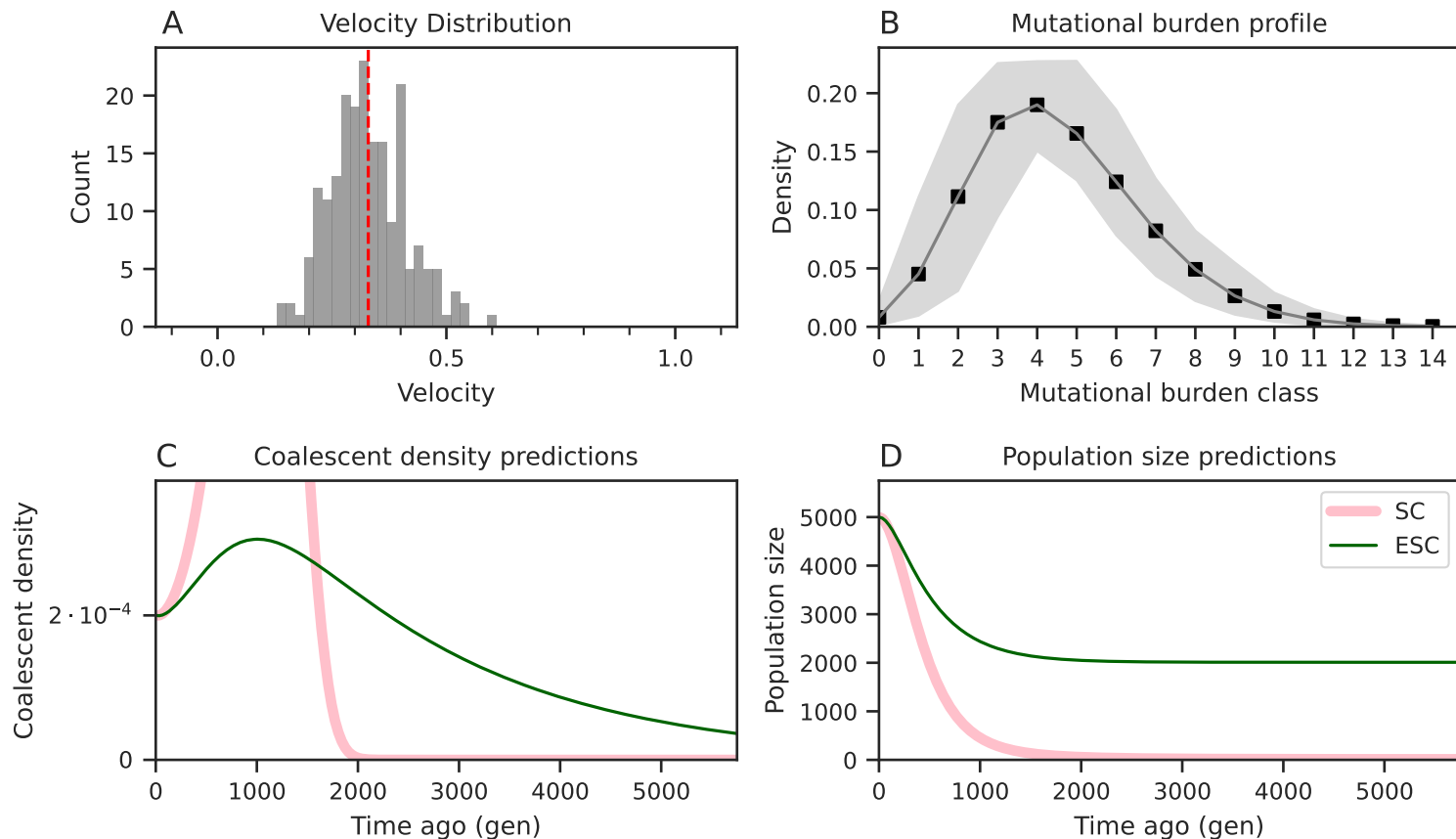
D Population size predictions



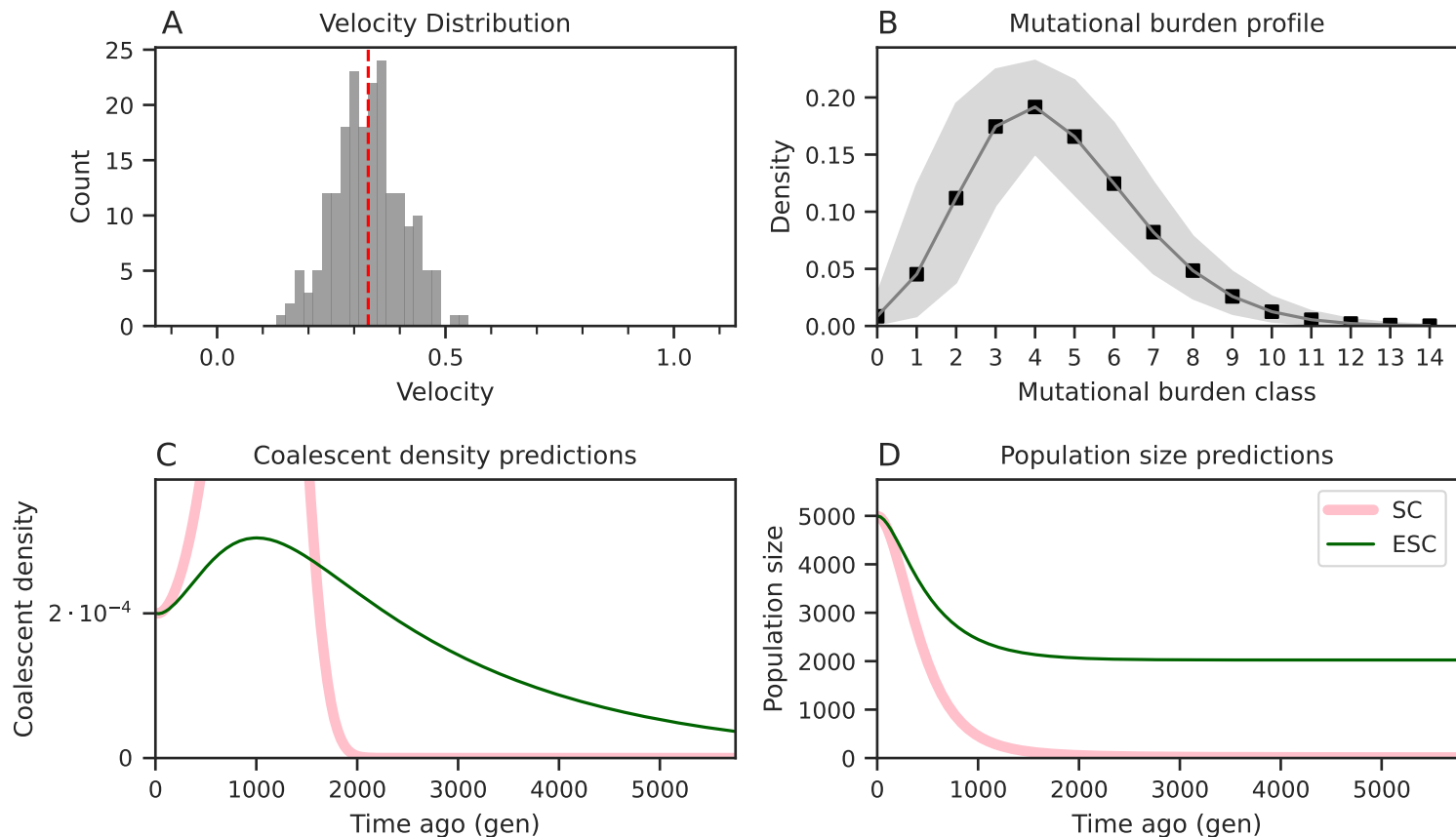
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 2.36 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity = 0.3257 ± 0.0817 , $n = 200$ simulations



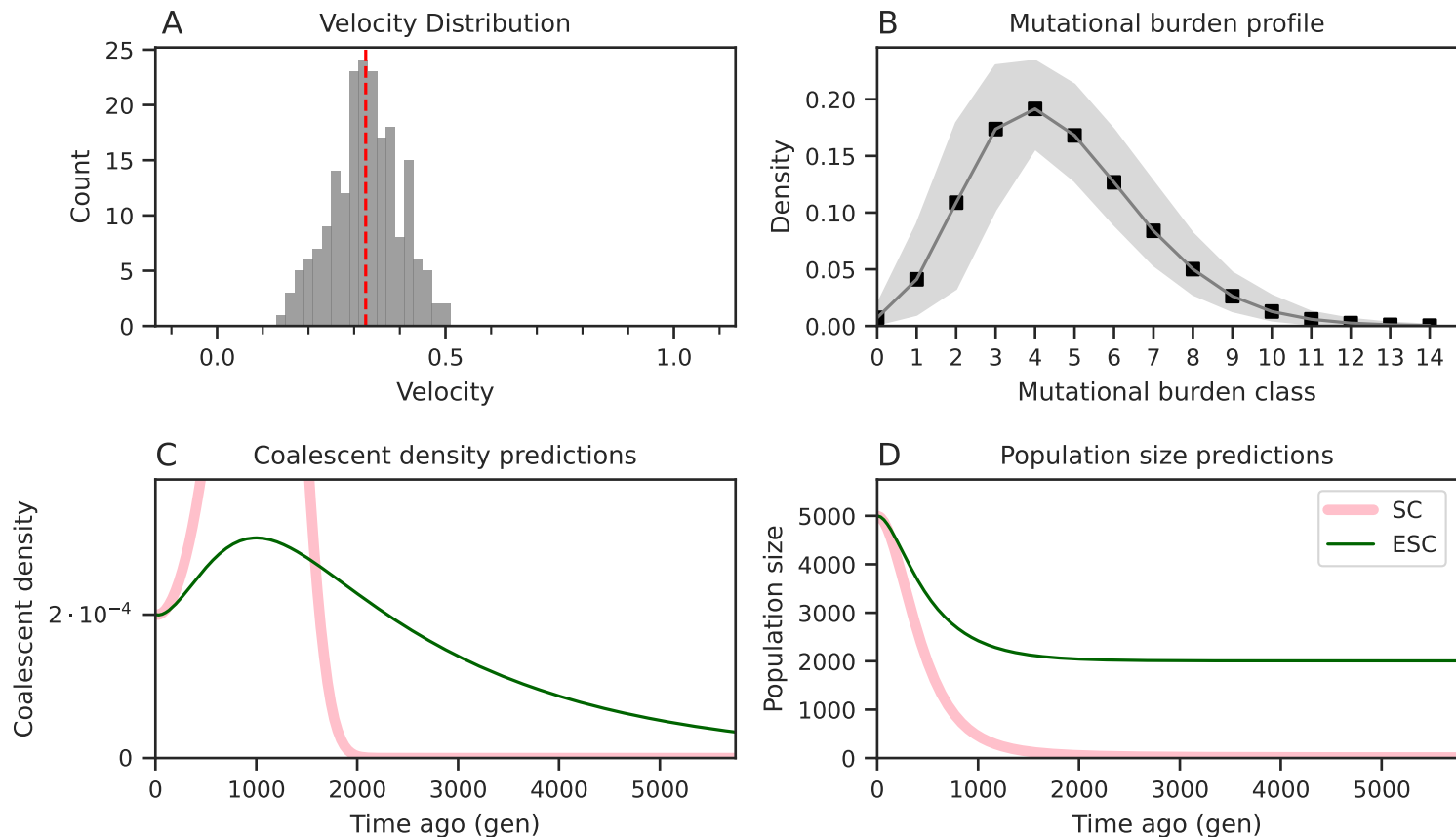
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 2.23 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity = 0.3294 ± 0.0823 , $n = 200$ simulations



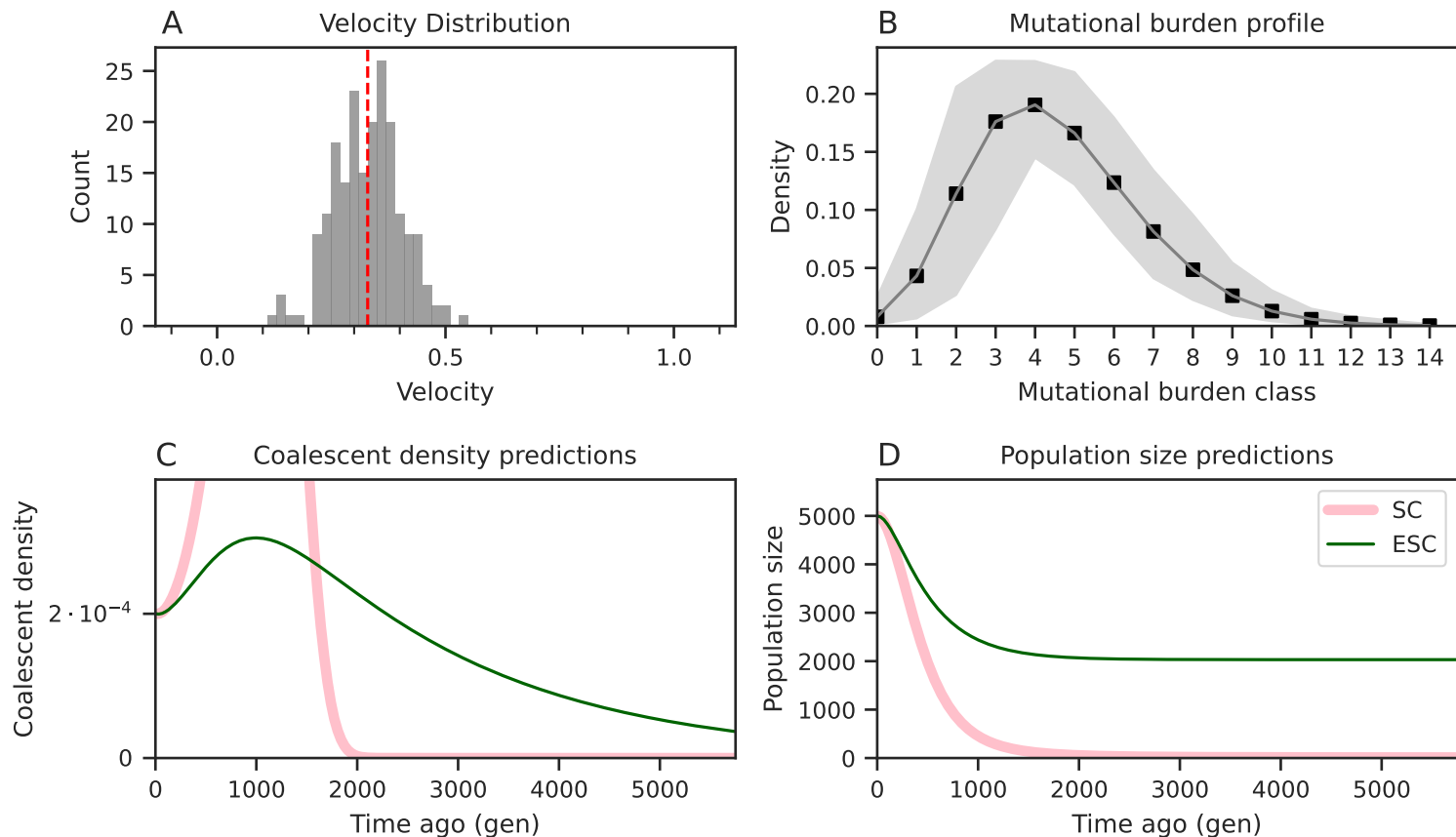
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 2.09 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3309 ± 0.0751 , $n = 200$ simulations



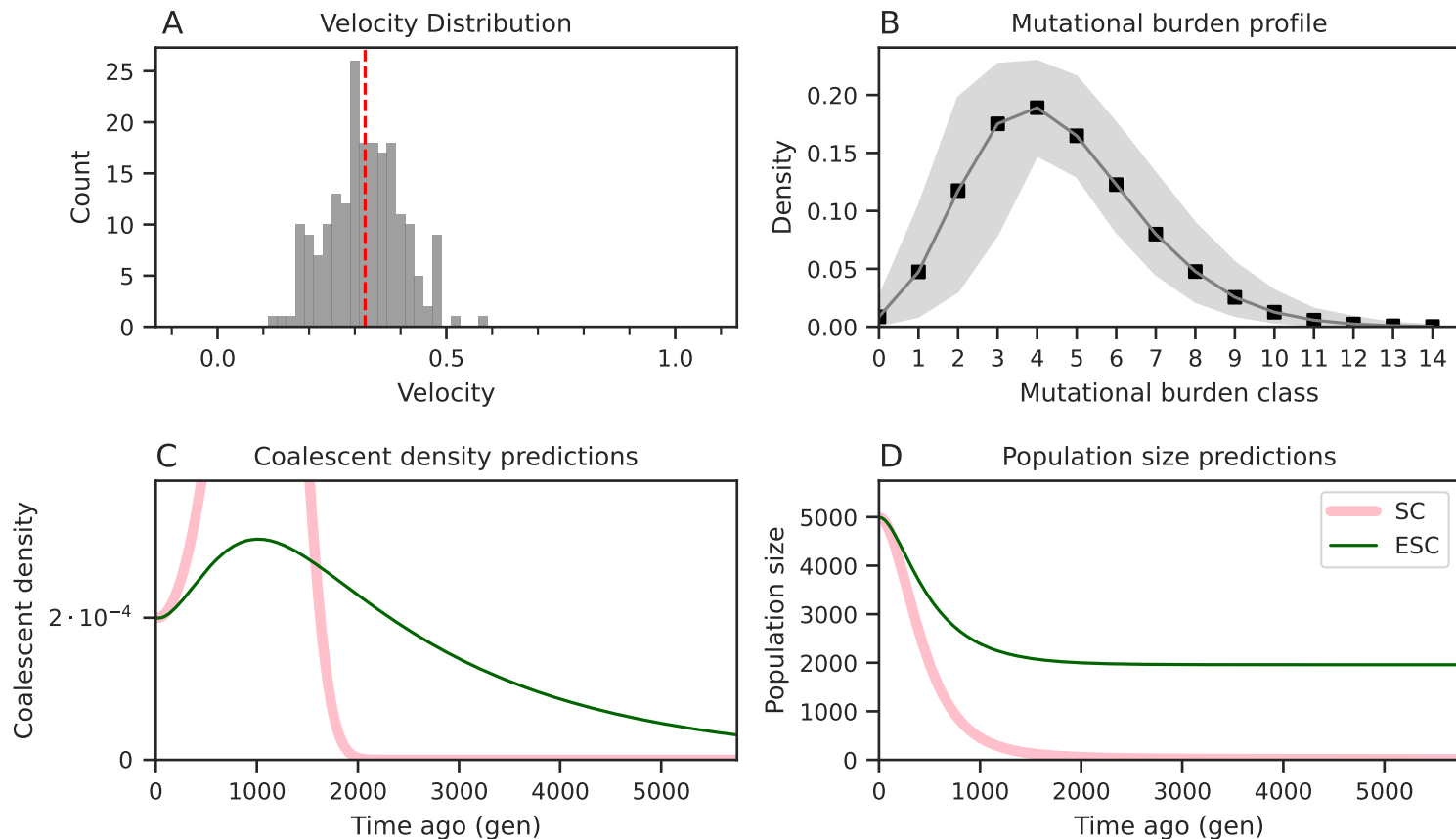
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 1.95 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity = 0.3254 ± 0.0733 , $n = 200$ simulations



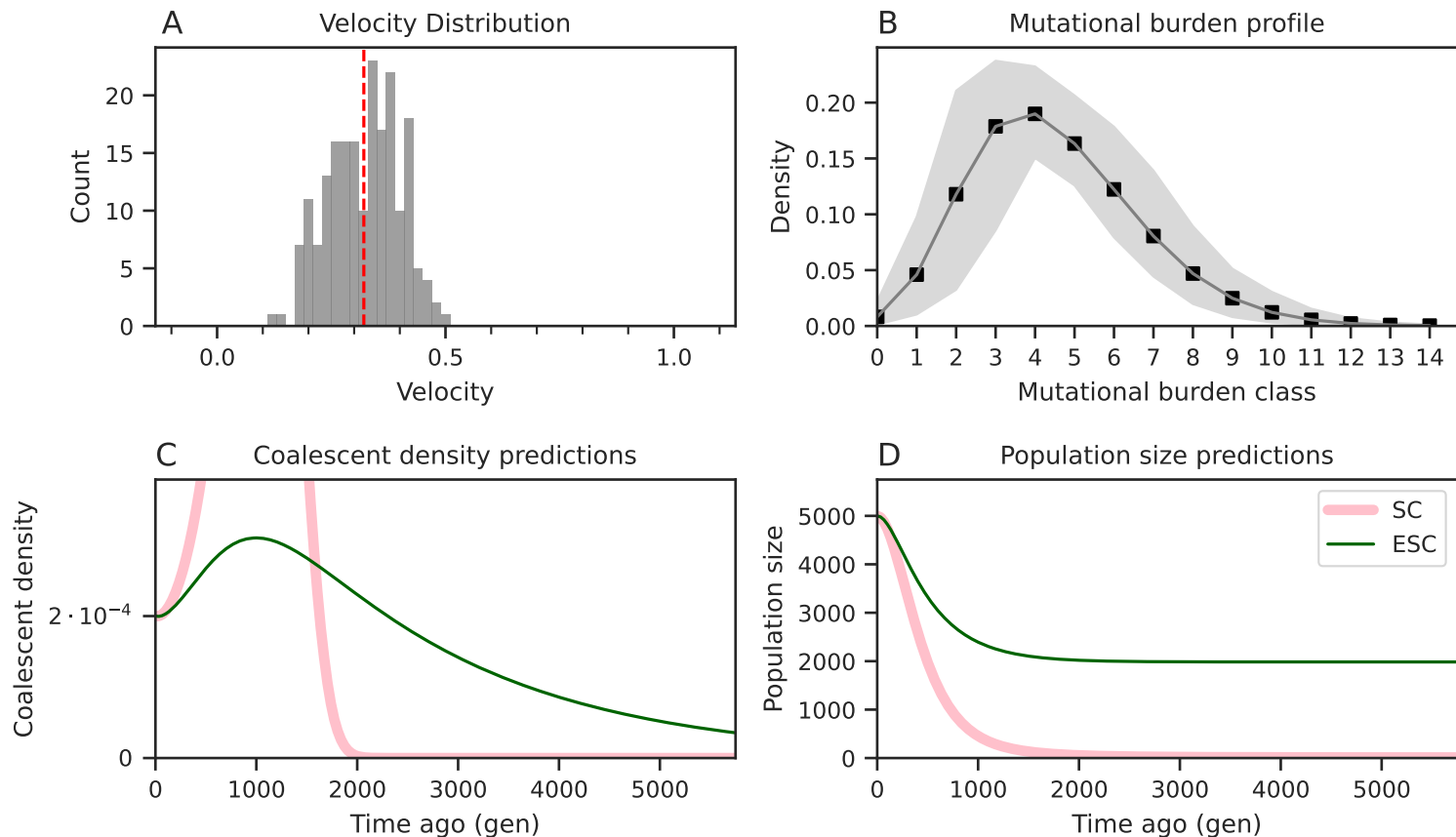
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 1.82 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3296 ± 0.0727 , $n = 200$ simulations



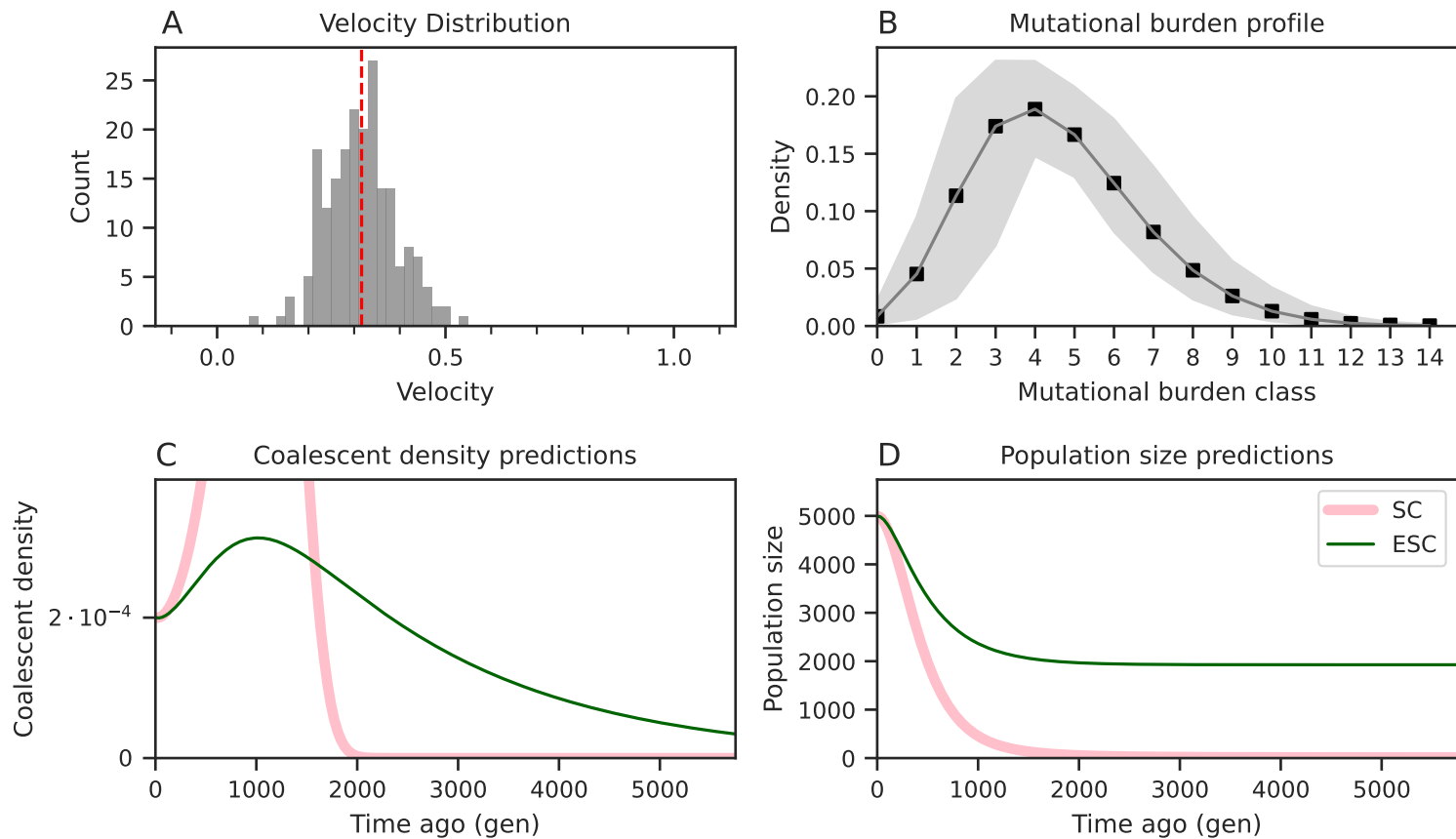
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 1.68 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3224 ± 0.0816 , $n = 200$ simulations



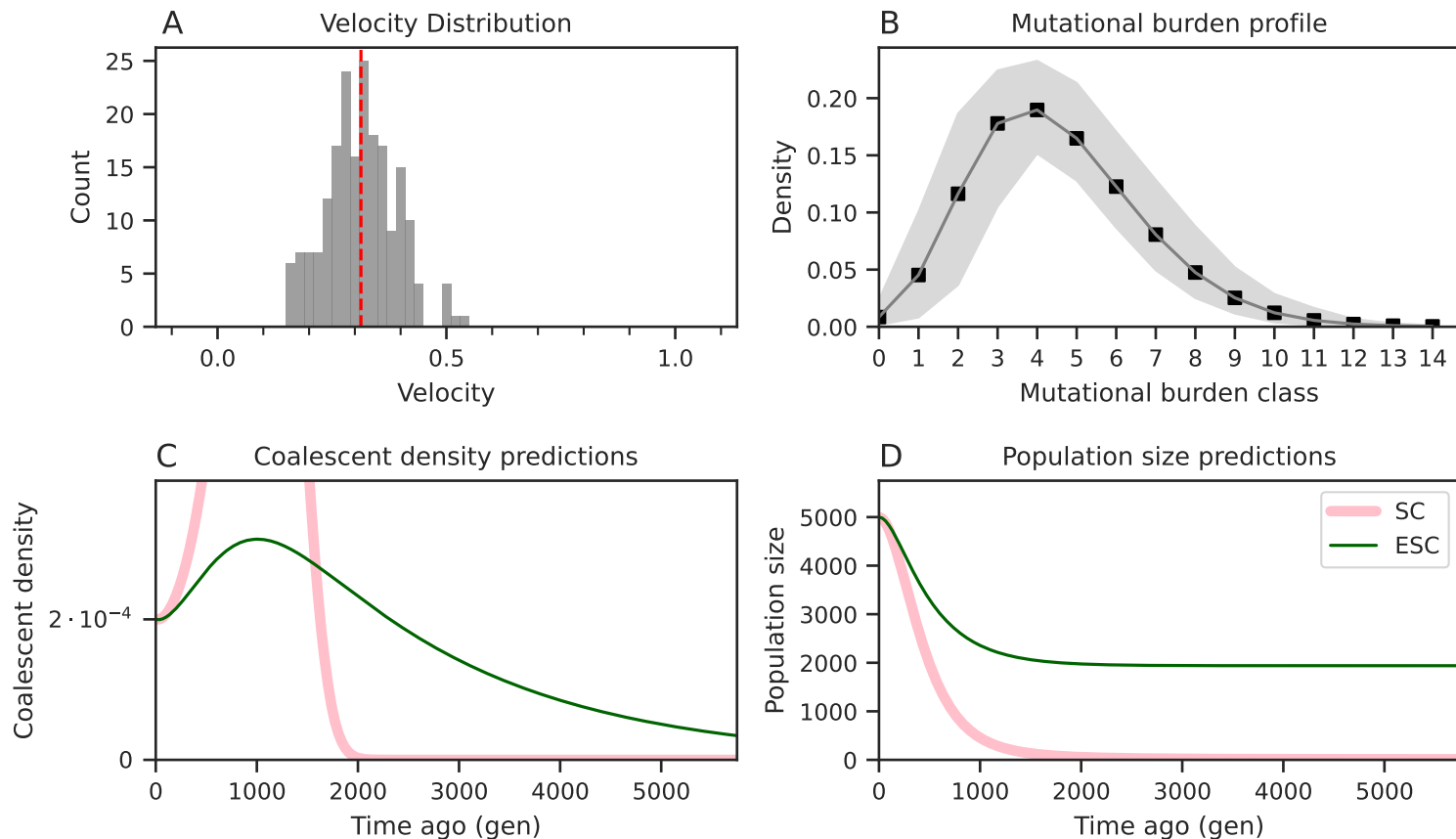
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 1.55 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity = 0.3212 ± 0.0774 , $n = 200$ simulations



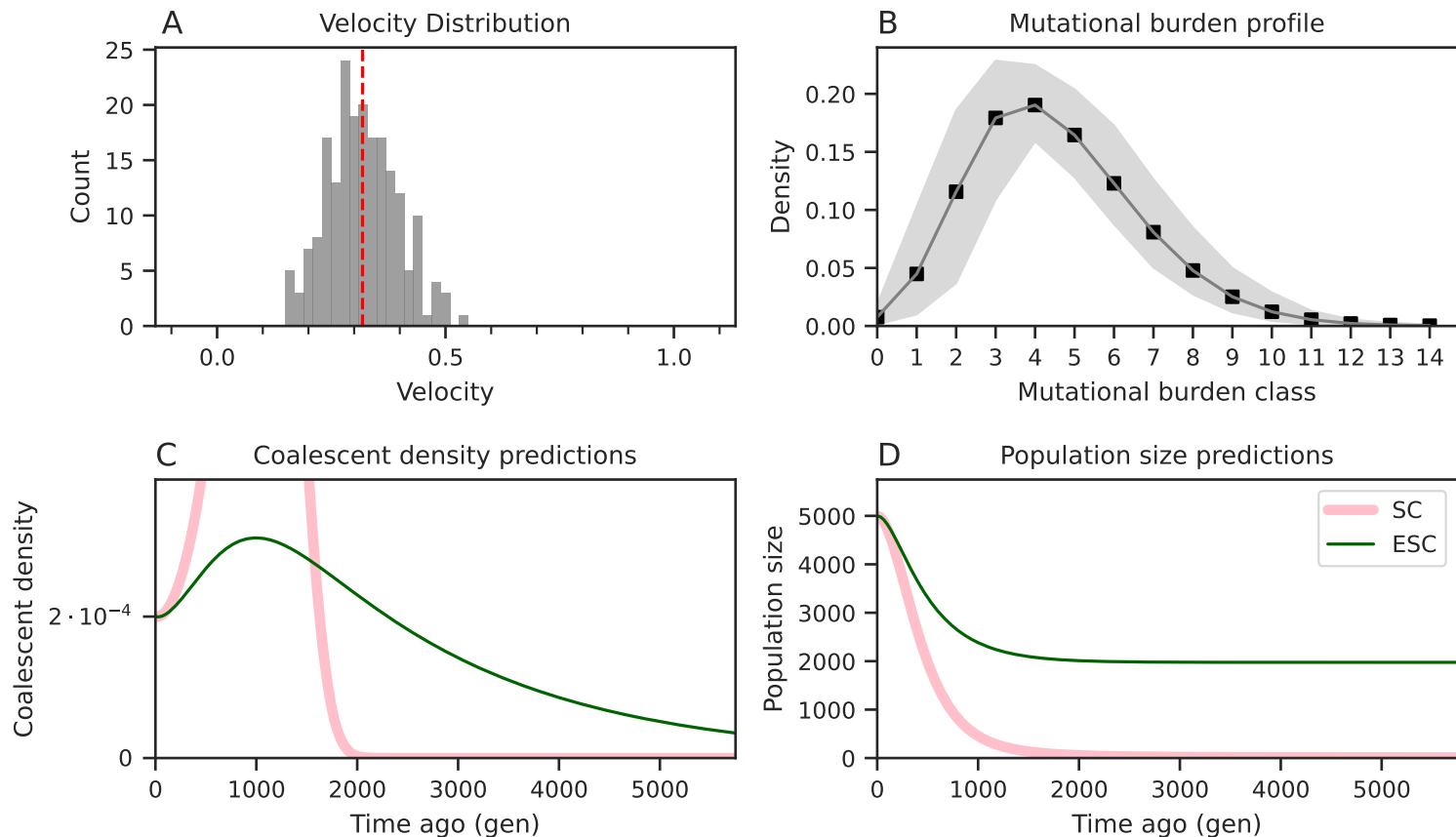
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 1.41 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3163 ± 0.0745 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 1.27 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3133 ± 0.0767 , $n = 200$ simulations

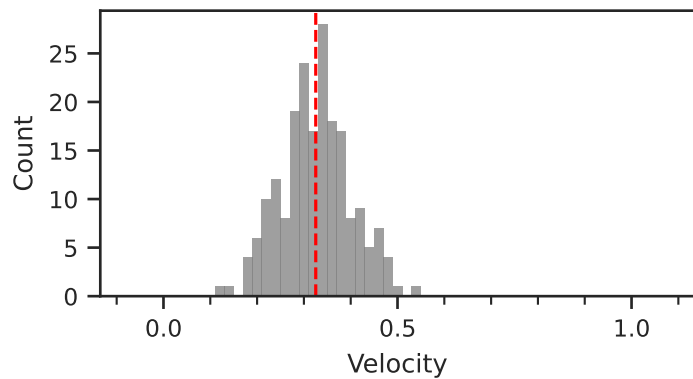


$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 1.14 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3183 ± 0.0773 , $n = 200$ simulations

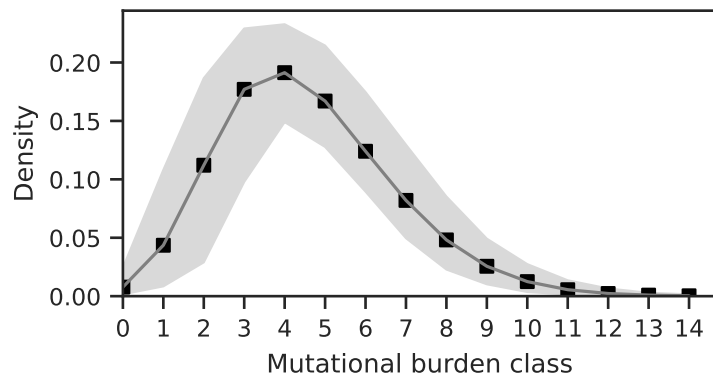


$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3253 ± 0.0741 , $n = 200$ simulations

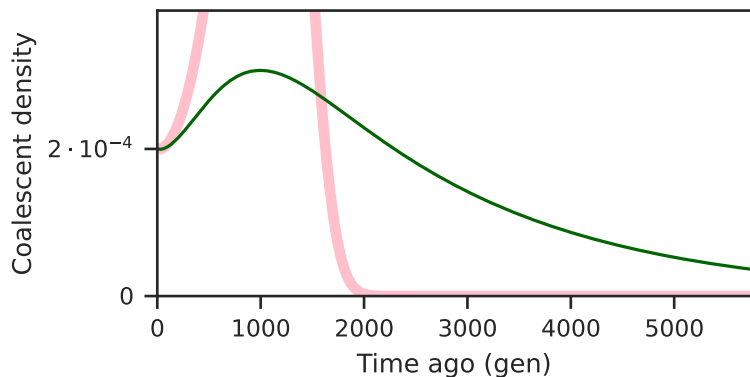
A Velocity Distribution



B Mutational burden profile



C Coalescent density predictions



D Population size predictions

